

LAB 6 – DELEGATES AND EVENTS

1. Basic delegate: add two numbers

```
delegate int MathOperation(int a, int b);
```

```
class Program
{
    static int Add(int a, int b) => (a + b);

    public static void Main(string[] args)
    {
        MathOperation m1 = Add;
        int result = m1(10, 20);
        Console.WriteLine($"sum = {result}");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab6$ dotnet run
sum = 30
```

2. Delegate subtract two numbers

```
delegate int MathOperation(int a, int b);
```

```
class Program
{
    static int Subtract(int a, int b) => (a - b);

    public static void Main(string[] args)
    {
        MathOperation m1 = Subtract;
        int result = m1(50, 20);
        Console.WriteLine($"difference = {result}");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab6$ dotnet run
difference = 30
```

3. Simple event: button “click”

```
class Button
{
    public delegate void ClickHandler();
    public event ClickHandler? Clicked;

    public void Click()
    {
        Console.WriteLine("Button is Clicked");
        Clicked?.Invoke();
    }
}
```

```
class Program
{
```

```

static void Button_Clicked()
{
    Console.WriteLine("Event Handler: Button clicked is received !");
}

```

```

public static void Main(string[] args)
{
    Button button = new Button();
    button.Clicked += Button_Clicked;
    button.Click();
    button.Click();
    button.Click();
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab6$ dotnet run
Button is Clicked
Event Handler: Button clicked is received !
Button is Clicked
Event Handler: Button clicked is received !
Button is Clicked
Event Handler: Button clicked is received !

```

4. Simple event : Alarm

```

class Alarm
{
    public delegate void ClickHandler();
    public event ClickHandler? Clicked;

    public void Ring()
    {
        Console.WriteLine("Alarm is ringing !");
        Clicked?.Invoke();
    }
}

class Program
{
    static void AlarmRang()
    {
        Console.WriteLine("Event Handler: Alarm is ringing !");
    }

    public static void Main(string[] args)
    {
        Alarm alarm = new Alarm();
        alarm.Clicked += AlarmRang;
        alarm.Ring();
        alarm.Ring();
    }
}

```

```

    }
}
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab6$ dotnet run
Alarm is ringing !
Event Handler: Alarm is ringing !
Alarm is ringing !
Event Handler: Alarm is ringing !

```

5. Synchronous method

```

class Program
{
    public static void Main(string[] args)
    {
        Console.WriteLine("Video is uploading!");
        uploadVideo();
        playVideo();
    }

    static void uploadVideo()
    {
        Thread.Sleep(10000);
        Console.WriteLine("Video is uploaded successfully!");
    }
    static void playVideo()
    {
        Console.WriteLine("Playing next video!");
    }
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab6$ dotnet run
Video is uploading!
Video is uploaded successfully!
Playing next video!

```

6. Asynchronous using async and await with Task.Delay()

```

class Program
{
    public static async Task Main(string[] args)
    {
        Console.WriteLine("Video is uploading!");

        await UploadVideoAsync();

        PlayVideo();
    }

    static async Task UploadVideoAsync()
    {
        await Task.Delay(10000);
        Console.WriteLine("Video is uploaded successfully!");
    }
}

```

```

        static void PlayVideo()
        {
            Console.WriteLine("Playing next video!");
        }
    }

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab7$ dotnet run
Video is uploading!
Video is uploaded successfully!
Playing next video!

```

7. Asynchronous using Task class

class Program

```

{
    public static async Task Main(string[] args)
    {
        Console.WriteLine("Video is uploading!");
        Task uploadTask = UploadVideoAsync();
        PlayVideo();
        await uploadTask;
    }
    static async Task UploadVideoAsync()
    {
        await Task.Delay(10000);
        Console.WriteLine("Video is uploaded successfully!");
    }
    static void PlayVideo()
    {
        Console.WriteLine("Playing next video!");
    }
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab7$ dotnet run
Video is uploading!
Playing next video!
Video is uploaded successfully!

```